Temperature-Barrier Physics: Vapor, Heat Capacity, and Ionization

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Plasmonic-Enhanced Laser-Induced Sonofusion: Overcoming the Temperature Barrier via Resonant Nanoparticle Heating

Abstract

I examine how temperature-dependent ideal-gas heat capacities and a fixed water-vapor inventory change the peak temperature of an acoustically driven bubble. Identical heat-transfer parameters separate caloric effects from transport changes; one-way post-processing separates those effects from mechanical feedback. The preceding constant-capacity model is recovered as an exact equation limit. Molecular excitation changes the air-like caloric response, whereas ground-state ideal argon retains constant heat capacity. Water vapor enters both energy and pressure, replacing the preceding separate constant vapor-pressure term. Chemical dissociation and ionization are assessed quantitatively as energy demands, without computing chemical populations or a chemically corrected peak. Coefficient boundaries censor a trajectory rather than impose a physical temperature ceiling. All extreme 1.3 bar cases remain available, subject to their numerical scope and strong physical limitations.

1 Objective and physical motivation

The question is how water vapor, variable heat capacity and endothermic processes modify compression heating, without assuming that every added mechanism lowers the resulting maximum. Increasing heat capacity reduces temperature gain for a specified energy input, but changing pressure changes the bubble trajectory, compression work and thermal history. Heat can enter the gas during expansion, so a comparison over an entire driven cycle is not an isolated adiabatic compression.

Vapor trapping and molecular excitation are established mechanisms in sonoluminescence models. Toegel et al. [1] couple vapor exchange, heat transfer and vibrational energy. Yuan [2] resolves a reacting multicomponent gas, while Hammer and Frommhold [3] include chemistry, ionization and radiation. Their closures, operating conditions and observables differ from this study. Their temperatures are not calibration targets. The local studies [9, 10, 11] likewise motivate the problem without validating a uniform gas temperature in water under the present conditions.

2 Reference state and controlled model hierarchy

The inherited conditions are $R_0 = 6\,\mu\text{m}$, $f = 30\,\text{kHz}$, $T_0 = T_\ell = 293.15$ K and a five-cycle requested interval $0 \le t \le 5/f = 166.667\,\mu\text{s}$. Acoustic amplitude is the peak sinusoidal amplitude, with

$$p_\infty(t) = p_0 - P_a \sin(2\pi f t), \qquad p_{d0} = p_0 + 2\sigma/R_0 - p_v = 123232.667 \text{ Pa}.$$

The water constants and forcing phase are unchanged (Appendix A). The air-like gas is a frozen 79/21 mole-percent N_2/O_2 idealization; the argon-like gas is ground-state Ar. No constituent reacts in the ODE.

- **Model 0:** the Mini-Research 3 limit, $c_V = R_u/(\gamma - 1)$, $\gamma = 1.4$ or 5/3, dry inventory $n_d = p_{d0}V_0/(R_uT_0)$, plus separate mechanical p_v.
- **Model 1:** the same legacy pressure convention and dry inventory, with NASA heat capacities; a same-component control freezes each heat capacity at T_0.
- **Model 2:** a closed ideal mixture with consistent vapor pressure and energy, including dry, initially saturated and enhanced trapped-vapor inventories. A same-mixture frozen-capacity control is retained.
- **Model 3 assessment:** chemical energy scales and hypothetical complete-conversion demands; this level supplies no new dynamical temperature prediction.

3 Thermodynamically consistent formulation

For fixed composition, define sensible energy relative to the initial state,

$$\Delta U(T) = \sum_i n_i \int_{T_0}^{T} [c_{P,i}(\theta) - R_u], d\theta, \qquad C_V^{\mathrm{tot}}(T) = \frac{dU}{dT} > 0, \tag{1}$$

$$p_g = \frac{n_{\mathrm{tot}} R_u T}{V}, \qquad V = \frac{4\pi R^3}{3}, \qquad \dot{V} = 4\pi R^2 u, \quad u = \dot{R}. \tag{2}$$

NASA9 data [4] give $c_P/R_u = a_1/T^2 + a_2/T + a_3 + a_4 T + a_5 T^2 + a_6 T^3 + a_7 T^4$. I integrate c_V analytically on each interval and add matching offsets, so energy is continuous at every join. The source enthalpy integration constants cancel for unchanged composition; no formation energy is counted twice. The model is

$$\boxed{\dot{T} = \frac{-p_g \dot{V} - \dot{Q}_{\mathrm{loss}}}{C_V^{\mathrm{tot}}(T)}}, \qquad \dot{Q}_{\mathrm{loss}} = 4\pi\eta k_g R(T - T_\ell), \qquad \dot{Q}_{\mathrm{source}} = 0. \tag{3}$$

The signed flux is negative when the liquid heats the gas. The retained k_g is 0.0264 W/(m K) for air and 0.0177 W/(m K) for argon [7], including their wet-mixture controls. These are fixed 300 K reference parameters, not a high-temperature mixture transport law.

The energy-consistent pressure derivative is

$$\dot{p}_g = p_g \left(\frac{\dot{T}}{T} - \frac{3u}{R} \right). \tag{4}$$

Let $p_c = p_v$ for legacy Models 0/1, and $p_c = 0$ for Model 2. The inherited Keller–Miksis (KM) convention is preserved:

$$\Delta p = p_g + p_c - p_\infty - 2\sigma/R - 4\mu u/R, \tag{5}$$

$$D\dot{u} = (1 + u/c)\Delta p/\rho + RB/(\rho c) - \tfrac{3}{2}(1 - u/(3c))u^2, \tag{6}$$

$$D = R(1 - u/c) + 4\mu/(\rho c), \tag{7}$$

$$B = \dot{p}_g + 2\sigma u/R^2 + P_a \omega \cos \omega t + 4\mu u^2/R^2. \tag{8}$$

The viscous acceleration derivative is included in D. A varying diagnostic $\gamma(T) = 1 + R_u/c_V(T)$ does not justify $pV^{\gamma(T)} = \text{constant}$. For this fixed mixture, the adiabatic invariant would instead involve $\int c_V(T)dT/T + R_u \ln V$.

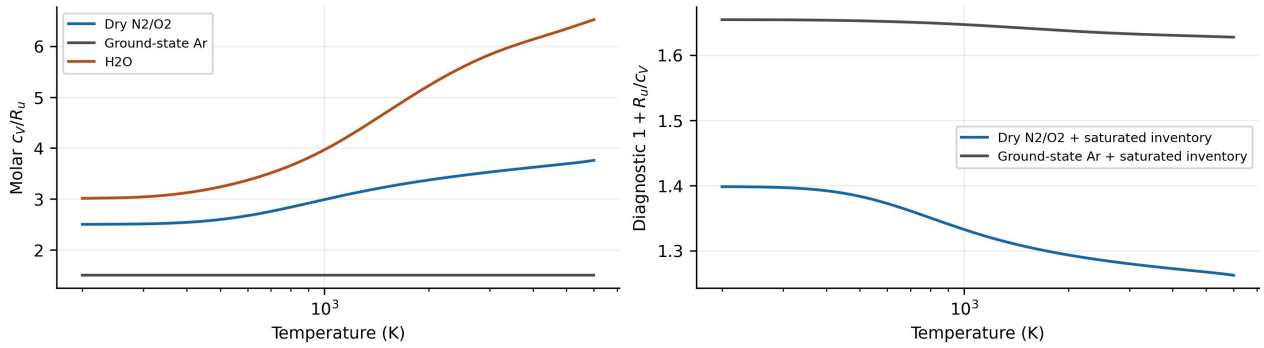


Figure 1: Per-mole heat capacity and a diagnostic capacity ratio. Molecular internal excitation is included in NASA species data. Ground-state argon remains calorically perfect. The ratio is not imposed as a variable-exponent pressure law.

NASA fits cover 200–20000 K for N_2/O_2 and 200–6000 K for H_2O. An event stops the accepted variable-capacity trajectory at its first domain boundary. Ground-state monatomic Ar has an analytic constant capacity; retaining that mathematical model above atomic excitation/ionization scales does not establish its physical validity.

4 Vapor inventory and numerical methodology

The saturated-inventory Model 2 uses

$$n_{v0} = \frac{p_v V_0}{R_u T_0}, \qquad n_{d0} = \frac{p_{d0} V_0}{R_u T_0}, \qquad p_{g0}^{\text{mixture}} = p_0 + 2\sigma/R_0.$$

Its water mole fraction is 1.8627%. Both pressure and energy contain this same inventory, with no additional constant p_v. Evaporation, condensation and latent heat are absent. For sensitivity, $x_v = b\,p_v/(p_0 + 2\sigma/R_0)$ with $b = 0, 1, 5, 10$, holding total initial pressure and total moles fixed by displacing dry gas. The enhanced cases are mechanically equilibrated hypothetical trapped inventories, supersaturated at T_0, rather than thermodynamic initial equilibria or a calculation of vapor capture. The dry Model 2 control isolates replacing the legacy vapor-pressure convention before adding water heat capacity.

The vapor-free Model 2 control replaces the initial vapor pressure by additional dry-gas moles to preserve static balance. Its contrast with Model 1 therefore includes this inventory normalization, as well as removal of constant p_v. The saturated case restores exactly the preceding dry inventory and adds the prescribed H_2O inventory.

One-way calculations reuse the same published mechanical trajectories and the same C1 Hermite radius reconstruction used by Mini-Research 3. These histories originate in Mini-Research 2; I do not substitute the newly coupled history into that comparison. The reconstructed velocity is exactly dR/dt. Thermodynamic pressure can differ from the pressure that generated the history; the maximum relative mismatch and its value at the thermal peak are saved. Coupled runs evolve radius, velocity, temperature and energy ledgers together.

Production uses adaptive DOP853, relative tolerance 2×10^{-9} and maximum step $(1/f)/600$. Refinement uses 2×10^{-11} and $(1/f)/1200$; independent Radau uses 2×10^{-10}. Temperature extrema are roots of Eq. (3) on dense polynomials, with endpoints included and a subdivided accepted-step search grid. One-way searches additionally include the mechanical knots. Radius minima are found independently. The saved thermal peak time, cycle, radius and pressure are not replaced by the radius-minimum event.

Table 1: Principal comparison at $\eta = 1$. OW: one-way; CO: coupled. M1 uses variable dry heat capacity and legacy p_v; M2 uses the fixed saturated inventory. A dagger denotes the maximum on an observed, censored interval, not a five-cycle global maximum. The relative change uses M2 versus the same-mode MR3 peak only when M2 completes five cycles.

P_a (bar)	Gas	Mode	MR3 (K)	M1 (K)	M2 (K)	$100\delta_T$
1.00	Air	OW	441.3	439.8	440.8	-0.1
1.00	Air	CO	707.1	693.4	552.9	-21.8
1.00	Ar	OW	576.0	576.0	574.0	-0.3
1.00	Ar	CO	886.8	886.8	661.4	-25.4
1.15	Air	OW	3898.7	2667.2	2647.1	-32.1
1.15	Air	CO	4423.4	3417.0	2798.4	-36.7
1.15	Ar	OW	22142.1	22142.1	6000.0†	–
1.15	Ar	CO	7893.0	7893.0	5674.9	-28.1
1.30	Air	OW	15174.0	6777.5	6000.0†	–
1.30	Air	CO	10555.0	6741.2	6000.0†	–
1.30	Ar	OW	213181.6	213180.8	6000.0†	–
1.30	Ar	CO	26421.6	26421.6	6000.0†	–

The complete design contains 92 runs: 76 complete their requested interval and 16 retain a censored interval. Every saved boundary event includes time, temperature and reason. The search is finite and does not establish an absolute maximum over all physical conditions.

5 Results and separation of mechanisms

At 1.3 bar the variable-capacity dry-air peaks are 6777.5 K (one-way) and 6741.2 K (coupled); the saturated-inventory observed peaks are 6000.0 and 6000.0 K. Corresponding argon values are 213180.8, 26421.6, 6000.0 and 6000.0 K. The scope markers in the principal table determine which of these are global over five cycles. A value at a coefficient boundary is not a prediction that the bubble cannot get hotter.

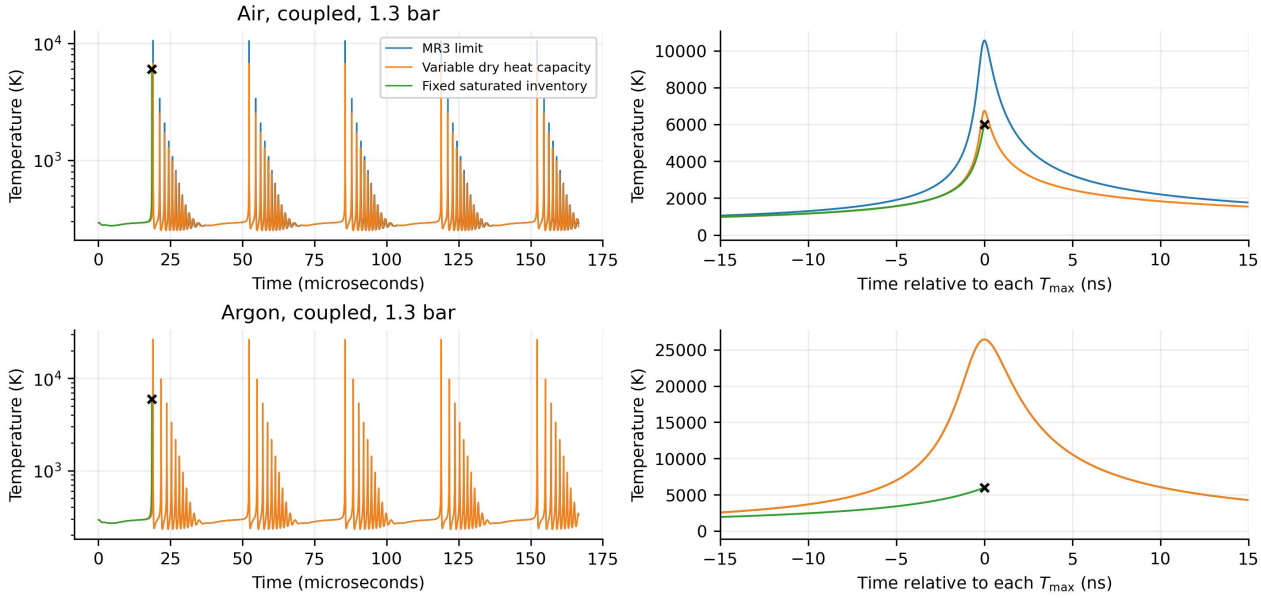


Figure 2: Coupled temperature histories at 1.3 bar. Crosses mark coefficient-domain termination where present. Right panels align each curve to its own refined thermal maximum, exposing narrow heating intervals; this alignment does not assert simultaneous collapses.

The variable-capacity effect is the ratio of Model 1 to its same-component frozen-$c_V(T_0)$ control, not to an arbitrarily chosen constant γ. This removes the small initial-capacity difference between the NASA N_2/O_2 mixture and idealized $\gamma = 1.4$. The ground-state Ar controls coincide within numerical precision. Its large dry temperatures therefore remain in the dataset.

In the wet argon comparison, variable versus frozen mixture capacity isolates H_2O excitation because Ar capacity is unchanged. For wet air the same contrast changes both molecular dry-gas and water capacities, and is labeled accordingly.

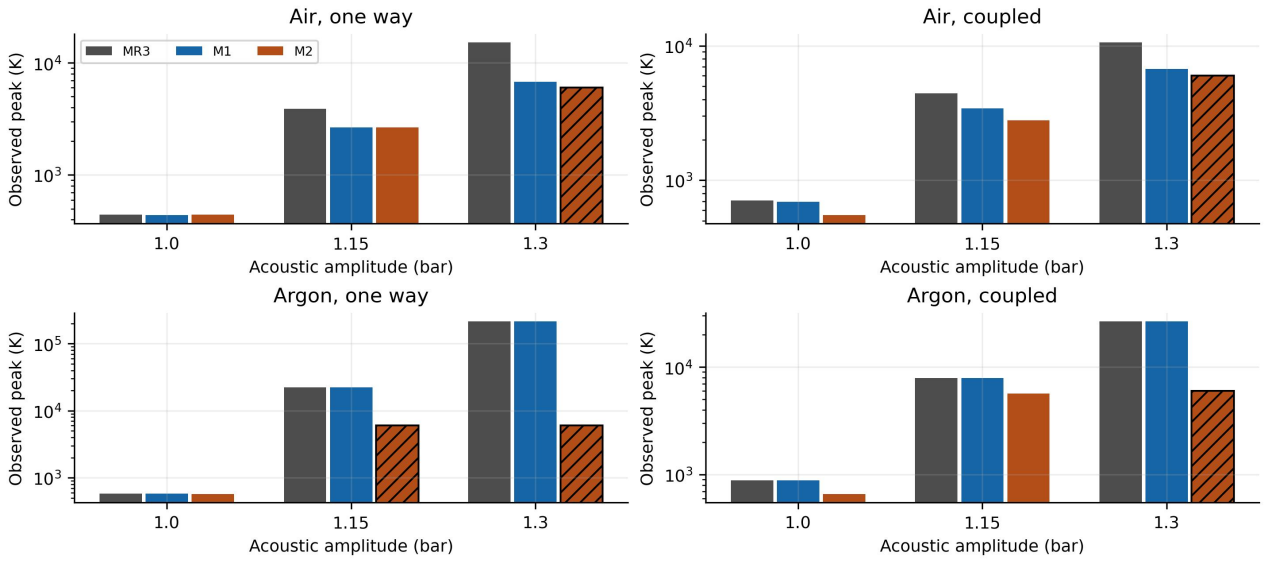


Figure 3: Saved peak comparisons. Hatched bars are censored observed peaks. The change from M1 to M2 includes both the vapor-pressure convention and the fixed water inventory, so it is not attributed solely to molecular heat capacity.

6 Sensitivity and energy budgets

The paired sensitivity table records the exact reference run for every change: variable versus frozen capacity, replacement of the vapor-pressure convention, water inventory at unchanged total initial pressure, and $\eta = 0.5, 1, 2$ at 1.3 bar and for air at 1.15 bar. The definition is $\delta_T = (T_{\max}^{\text{new}} - T_{\max}^{\text{reference}})/T_{\max}^{\text{reference}}$. Global changes are left undefined when either run is censored; separately labeled observed-maximum ratios are only interval diagnostics. One-way and coupled effects remain separate.

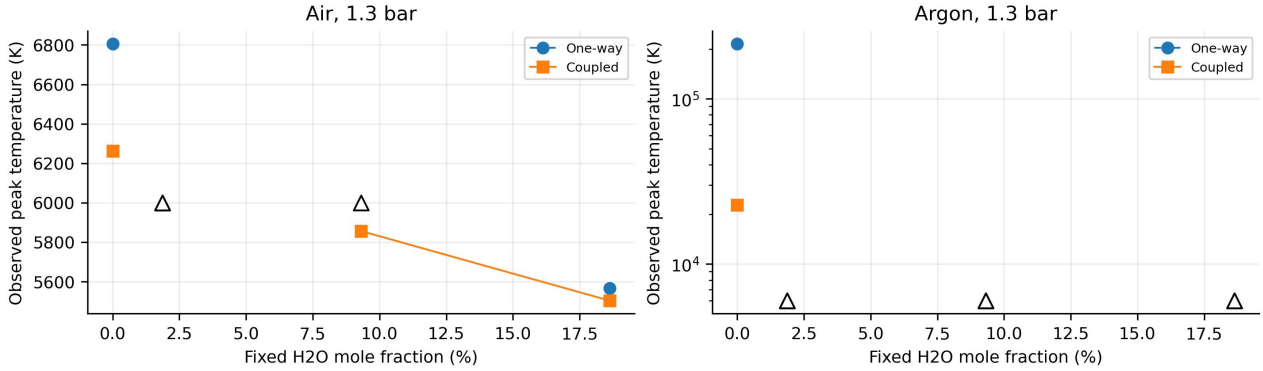


Figure 4: Discrete trapped-vapor scenarios at 1.3 bar. Open triangles denote coefficient-limited observed intervals; their continuation is uncomputed. Only complete points are connected. Fractions above saturation are conditional fixed-inventory states.

For the closed gas subsystem the checked identity is

$$\boxed{\Delta U(t) + W_{\text{by}}(t) + Q_{\text{loss}}(t) = 0}, \quad W_{\text{by}} = \int_0^t p_g \dot{V}, dt, \quad Q_{\text{loss}} = \int_0^t \dot{Q}_{\text{loss}}\, dt. \tag{9}$$

Gross compression $\int_0^t \max(-p_g \dot{V}, 0) dt$ is also retained, including the last expansion maximum to the thermal peak. It differs from signed net work. Chemical energy change is zero in the frozen calculations. Neither this identity nor a small residual proves conservation of energy in the entire gas–liquid system: the liquid is a fixed reservoir, and legacy p_v has no vapor energy ledger.

Table 2: Saturated-inventory energy at the refined observed thermal peak. W_{by} is work done by the gas; heat loss is signed. ϵ_E is the maximum residual over the observed interval, normalized by the largest sensible-energy, work, heat or initial nR_uT_0 scale.

P_a	Gas	Mode	ΔU (nJ)	W_{by} (nJ)	Q_{loss} (nJ)	ϵ_E
1.00	air	OW	0.1453	-0.1811	0.0357	8.7e-10
1.00	air	CO	0.2586	-1.1540	0.8955	2.0e-11
1.00	argon	OW	0.1667	-0.2563	0.0897	1.7e-10
1.00	argon	CO	0.2187	-0.2196	0.0009	2.1e-12
1.15	air	OW	2.8763	-2.7755	-0.1008	4.1e-07
1.15	air	CO	3.0836	-2.9584	-0.1252	1.3e-07
1.15	argon	OW	3.4821	-3.4593	-0.0228	6.0e-09
1.15	argon	CO	3.2814	-4.7057	1.4244	7.3e-08
1.30	air	OW	7.6641	-7.4619	-0.2022	8.4e-09
1.30	air	CO	7.6641	-7.4530	-0.2112	5.7e-09
1.30	argon	OW	3.4821	-3.3651	-0.1170	4.2e-09
1.30	argon	CO	3.4821	-3.3598	-0.1223	3.5e-09

The time-resolved energy decomposition is shown in Appendix A.

7 Dissociation and ionization: quantitative assessment

NIST formation enthalpies [5] give standard reaction energy costs. For dissociation I use $\Delta_r H^\circ(298.15\,\mathrm{K})$ and $\Delta_r U^\circ = \Delta_r H^\circ - \Delta\nu R_u T$. These are reference energy scales, not zero-temperature bond energies, kinetic barriers or a high-temperature equilibrium calculation. Ionization uses NIST isolated-atom ground-state thresholds [6] and exact SI conversion $1\,\mathrm{eV/particle} = 96485.3321\,\mathrm{J/mol}$.

Table 3: Energy scales. Dissociation "Energy" denotes $\Delta_r H^\circ_{298}$ in eV per parent molecule; "Cost" is $\Delta_r U^\circ_{298}$. Ionization entries use the ground-state threshold in both columns. The E/R_u scale is not an onset temperature.

Process	Energy (eV)	Cost (kJ/mol)	E/R_u (K)
N2 → 2N	9.798	942.88	113701
O2 → 2O	5.165	495.88	59939
H2O → OH + H	5.170	496.34	59994
H2O → 2H + O	9.608	922.04	111493
Ar → Ar+ + e^-	15.760	1520.57	182883
H → H+ + e^-	13.598	1312.05	157803
O → O+ + e^-	13.618	1313.94	158031
N → N+ + e^-	14.534	1402.33	168661

For each selected 1.3 bar inventory I calculate $E_{\mathrm{complete}} = n_{\mathrm{parent}}\,\Delta_r U^\circ_{298}$ or $n_{\mathrm{atom}} N_A E_i$, and compare it to gross compression, work during the selected collapse and the frozen sensible-energy increase. Atomic H/O/N ionization inventories are hypothetical inventories after complete dissociation, not actual neutral populations. First-step and full water dissociation are alternative channels; their demands must not be added. Reverse reactions would release chemical energy.

For the saturated inventory at the 1.3 bar coupled observation boundary, first-step water dissociation demands 0.431 nJ and full atomization 0.801 nJ. Complete N_2 dissociation in air would require 34.074 nJ (4.45 times its frozen sensible-energy increase). Complete first ionization of Ar would require 69.559 nJ, or 19.98 times its sensible-energy increase and 17.60 times its selected-collapse compression work. These demands quantify possible competition for energy, not realized conversion.

The saved inventory budgets show whether substantial chemical conversion would demand a small or large fraction of the frozen model's energy input. At a prescribed available energy E_{av} and fixed reference cost E_{complete}, energy accounting alone would permit at most a fraction $\min(1, E_{\mathrm{av}}/E_{\mathrm{complete}})$ if all other final energy demands were nonnegative and no further work or heat entered. This is a conditional bookkeeping restriction: reference costs, product sensible energy and the trajectory all change in a reacting bubble. It does not determine a realized reaction fraction or a temperature bound. Comparing $\exp[-E/(R_u T)]$ at 1000–30000 K supplies an energy-scale diagnostic, not a fraction or rate; density, translational entropy, partition sums, collision partners and kinetics are missing.

Vibrational and rotational contributions already present in the molecular heat capacities redistribute sensible energy internally. Dissociation adds chemical storage and changes the number of particles. Ionization additionally adds electrons and charged species, changing both pressure and energy. Subtracting a fixed conversion energy from the frozen curve would ignore those feedbacks, the temperature dependence of product energy and recombination. I therefore report no chemically corrected $T_{\max}$.

If an equilibrium closure were used, it would require a single $U(T,V)$ and $p(T,V)$ with

$$\dot{T} = \frac{-(p + U_V)\dot{V} - \dot{Q}_{\mathrm{loss}}}{U_T}, \qquad \dot{p} = p_T \dot{T} + p_V \dot{V}.$$

The present frozen mixture has $U_V = 0$ and $U_T = C_V^{\mathrm{tot}}$. A Saha calculation would additionally need nuclei conservation, charge neutrality, electron pressure/energy, statistical weights and partition sums, and evidence for local thermodynamic equilibrium (LTE). None of these is replaced by an energy subtraction.

8 Numerical evidence and physical applicability

The exact Model 0 equation limit is tested against the preceding implementation, including the pressure derivative and full viscous KM term. The largest observed regression peak difference is `3.71e-06`. For the checked principal calculations the largest production-to-strict peak difference is `3.48e-06`, and the independent-Radau difference is `1.82e-06`. Mechanical-input grid changes produce at most `4.91e-06`. These observed differences are convergence evidence, not rigorous error bounds. In weak 1.0 bar cases successive cycle peaks nearly tie: refinement can shift the selected argmax by one period despite a stable peak value. I retain the actual argmax and all candidate times, and report the span within a declared 10^{-8} relative near-peak band; this band is a conditioning diagnostic, not an error bound. Boundary-time convergence is checked separately from the uncomputed tail.

The maximum production energy residual is `4.38e-07`. Independently integrating work and heat on the refined dense histories with Gauss–Legendre quadrature gives a maximum 16-point scaled endpoint residual of `1.24e-06`. This independent quadrature checks the ledgers without equating a small ODE residual with full-system conservation. Numerical error, physical coefficient uncertainty and model inadequacy are separate: better tolerances do not correct fixed transport coefficients or an inappropriate EOS.

High pressures and densities challenge the ideal mixture, dilute thermodynamic data, continuum transport and a homogeneous gas temperature. Wall Mach number $|u|/c$ and $\chi = \max |p_L - p_0|/(\rho c^2)$ are saved for every run. The latter is a pressure-scale proxy, not a liquid density prediction. Strong liquid compressibility, shocks, finite molecular volume, interfacial transfer, radiation and shape instability are unresolved. Prescribed negative far-field liquid pressure during part of the strongest forcing represents modeled tension; it is not silently discarded or established as experimentally sustainable.

Thermal peak widths, the time from the preceding radius maximum to the thermal peak, and the ideal-gas acoustic crossing time $R/\sqrt{\gamma R_u T/M}$ are saved. These times can expose the weakness of assuming a single temperature, but do not provide vibrational or chemical relaxation rates. When relaxation is comparable to a narrow collapse pulse, equilibrium heat capacities and LTE chemistry need separate justification. Published finite-rate and spatial models [2, 3] demonstrate why an energy-only assessment cannot settle those issues.

Numerically reproducible. The specified equations, saved maxima, domain events, ledgers and paired comparisons are reproducible within the documented numerical differences.

Conditional model-based estimate. Every computed peak remains conditional on a spherical, uniform, ideal, frozen gas, prescribed heat transfer and the adopted KM liquid description. Incomplete runs have only observed-interval maxima.

Physically supported. Molecular internal degrees of freedom, signed thermal exchange, energy-consuming dissociation and nonzero atomic ionization energy are established mechanisms. That support does not validate the extreme numerical temperatures.

Requires a more complete physical model. Quantitative high-density peaks, actual vapor capture, reacting composition, shock structure and nonequilibrium plasma states require additional EOS, spatial transport and rate information.

9 Conclusions

Variable molecular heat capacity, vapor pressure/energy accounting and mechanical feedback are distinguishable contributions. Ground-state argon supplies a useful null control for variable heat capacity. A fixed vapor mixture supplies a quantitative closed-system comparison, while coefficient-boundary events preserve the limit of what its data justify. Chemical energy costs indicate mechanisms that can compete with compression energy; they do not supply a new temperature cap or a chemically equilibrated trajectory.

No extreme scenario is declared physically impossible on the basis of this reduced model, and no finite parameter set establishes a universal maximum. The study provides material for independent scientific assessment. It does not infer fusion, implement optical heating, or extend the model to subsequent research stages.

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A Parameters, model scope and stored quantities

The fixed liquid parameters are $p_0 = 101325$ Pa, $p_v = 2339$ Pa, $\sigma = 0.07274$ N/m, $\rho = 998.2$ kg/m^3, $\mu = 0.001002$ Pa s and $c = 1482$ m/s. They retain the preceding near-ambient water approximation referenced to IAPWS [8]; no claim is made that they remain locally constant at extreme collapse pressure. The universal gas constant is $R_u = 8.31446261815324$ J/(mol K). Molar masses are 0.0280134 (N_2), 0.0319988 (O_2), 0.01801528 (H_2O) and 0.039948 kg/mol (Ar). Mixture density is $\rho_g = \sum_i n_i M_i / V$.

Each trajectory includes temperature, radius, wall velocity, pressure, density, sensible energy, signed work and heat, gross compression, and the energy residual. The summary includes exact model/transport parameters, peak time/cycle, thermal and radius events, observed interval, integration completion and applicability diagnostics. Coefficients and their units/ranges are stored separately. The data retain every 1.3 bar scenario without a Mach-based exclusion or temperature clipping.

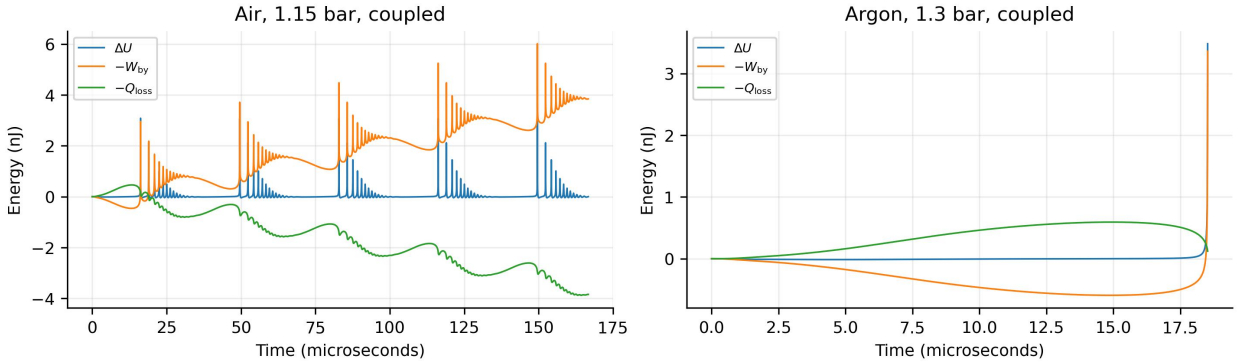


Figure 5: Energy decomposition from the saved trajectories. Sensible energy equals net compression input minus signed heat loss within the numerical residual. The 1.3 bar argon curve ends at the coefficient boundary. No chemical energy sink is imposed.

B Convergence details and reproducibility

Table 4: Selected saturated-mixture convergence comparisons against production, including censored cases. Time differences refer to the refined thermal peak; a one-period change in the weak 1.0 bar case selects a nearly tied cycle (Section 8). Full input-grid and Model 0/1 comparisons are retained in CSV.

Case	Check	$\Delta T/T$	Δt_T (ns)	Scope
air / 1.00 / OW	strict	-4.15e-07	4.1e-05	full
air / 1.00 / CO	strict	-2.79e-11	-3.33e+04	full
argon / 1.00 / OW	strict	6.49e-07	-3.28e-05	full
argon / 1.00 / CO	strict	-1.71e-10	4.25e-08	full
air / 1.15 / OW	strict	1.76e-06	-1.07e-09	full
air / 1.15 / CO	strict	5.61e-08	1.57e-07	full
argon / 1.15 / OW	strict	-3.50e-13	-1.64e-05	censored
argon / 1.15 / CO	strict	2.00e-10	1.11e-06	full
air / 1.30 / OW	strict	2.16e-12	-1.09e-06	censored
air / 1.30 / CO	strict	8.99e-14	-6.84e-08	censored
air / 1.00 / OW	Radau	-4.03e-07	3.97e-05	full
air / 1.00 / CO	Radau	-4.39e-12	-2.35e-08	full
argon / 1.00 / OW	Radau	6.85e-07	-3.46e-05	full
argon / 1.00 / CO	Radau	-2.29e-10	5.67e-08	full
air / 1.15 / OW	Radau	1.82e-06	-1.11e-09	full
air / 1.15 / CO	Radau	5.72e-08	1.65e-07	full
argon / 1.15 / OW	Radau	-9.26e-14	-2.39e-05	censored
argon / 1.15 / CO	Radau	-1.64e-09	4.16e-07	full
air / 1.30 / OW	Radau	2.29e-12	-1.25e-06	censored
air / 1.30 / CO	Radau	1.47e-12	-7.15e-08	censored
argon / 1.30 / OW	strict	-2.15e-13	-1.54e-05	censored
argon / 1.30 / CO	strict	-2.03e-13	-8.3e-08	censored
argon / 1.30 / OW	Radau	8.73e-14	-1.63e-05	censored
argon / 1.30 / CO	Radau	-3.94e-14	-1.21e-07	censored

Absolute tolerances are 10^{-16} m in radius, 10^{-8} m/s in velocity, 2×10^{-10} in T/T_0 and 10^{-8} in each energy ledger scaled by nR_uT_0. The same absolute tolerances accompany stricter relative tolerances. Technical radius/denominator guards are numerical domain protection, not physical collapse limits. Internal trial stages can evaluate the nearest heat-capacity polynomial outside its documented interval solely to locate the first boundary; no accepted output past that boundary is used.

Tables and figures are generated from saved numerical data, not independent hand-entered temperatures. The reproducibility instructions specify locked Python dependencies and separate calculation, validation, plotting and document-building steps. The package preserves source inventories, reaction inputs, all sensitivities, boundary records and independent quadrature. The read-only sibling studies supply the accepted mechanical paths and reference data. An independent assessment should examine the thermodynamic closure, censoring, reference contrasts and physical interpretation as well as rerun the numerical tests.